Debarshi Chakraborty

LinkedIn | GitHub | Portfolio

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Experience

- **Associate Consultant** — July 2026 – Present
  Ernst & Young LLP — *Kolkata, India*
  - Delivered an **agentic supply planner** for a global CPG client's Digital Supply Planner program (**40+ countries**), eliminating **6,500+ hours** of manual planning effort.
  - Designed a **multi-agent architecture** (**10 sub-agents**); authored and evaluated **Databricks Genie** agents.
  - Applied **Bayesian MCMC** and **PySpark** simulation to minimize deadstock and run what-if inventory scenarios.
  - Added **long-term user personalization** and **PPT generation** from any response or full conversation.
  - Productionized with **Databricks Asset Bundles** for versioned, repeatable deployments across environments, cutting manual agent setup time by **80–90%**, and exposed **API endpoints**.
  - Developed **demand forecasting** models (**Random Forest**, **XGBoost**, post-processing) on time-series data, owning the **MLOps lifecycle** from backtesting and parallel runs to **Azure** and **Databricks** deployment.
- **AI/ML Intern** — January 2026 – July 2026
  Rezolve Ai — *Kolkata, India*
  - Engineered a production **Agent-to-Agent (A2A)** multi-agent system end-to-end (orchestrator, **MCP** tool servers, **Redis**-based context sharing) powering a live retail conversational AI platform.
  - Developed a **LangMem**-powered long-term memory agent that reduced redundant user inputs by **25%**.
  - Refactored the prompt stack into a **3-layer architecture** in **Langfuse**, cutting prompt-related incidents by **30%** across **6+ sub-agents**.
  - Led **LLM** evaluation for tool calling; model selection held routing performance and lowered latency by over **40%**.

Projects

- **Mnemos: Agentic Memory System with Provenance-Tracked Consolidation** — github.com/debarshi29/mnemos
  - Created a self-hosted agentic memory system with a **LangGraph-orchestrated** "sleep cycle" that extracts, deduplicates, and contradiction-checks facts from episodic history.
  - Combined **decay-ranked retrieval**, LLM-based contradiction resolution, and **EMA confidence updates** to preserve permanent provenance chains.
  - Shipped memory operations as **MCP tools** for Claude Desktop/Cursor with a React inspection UI, and tested them against 20 multi-session eval scripts.
- **FinSight: Regulatory-Grade Multi-Agent Financial Intelligence System** — github.com/debarshi29/FinSight
  - Architected a **six-stage multi-agent RAG pipeline** (Planner, Retriever, Analyst, Auditor, Comparator, Synthesizer) as a **LangGraph StateGraph** with hybrid BM25 + dense retrieval and cross-encoder reranking.
  - Engineered an entailment-checking **Auditor agent** that hard-blocks unverifiable claims, plus a deterministic currency normalizer that eliminates LLM arithmetic hallucination.
  - Added a **Qdrant**-based cross-session memory layer kept outside the claims/citations path, preserving auditability.

Technical Skills

Generative AI & Agents: LLMs, Generative AI, RAG, Agentic AI, Multi-Agent Systems, MCP, Prompt Engineering, LLM Evaluation, Hybrid Search, LangGraph, LangChain, LangMem, Langfuse
ML & Statistics: PyTorch, Scikit-learn, XGBoost, Random Forest, Time-Series Forecasting, Bayesian MCMC, GNNs
Data & Cloud: Databricks, Unity Catalog, PySpark, Delta Lake, MLflow, Azure, Azure OpenAI, GCP, Pandas, NumPy
MLOps & Infrastructure: Docker, Git, GitHub, FastAPI, Redis, Qdrant, Weaviate, Neo4j
Languages: Python, SQL, R, Java, C++

Education

- **Ramakrishna Mission Vivekananda Educational and Research Institute** — July 2024 – July 2026
  Master of Science, Big Data Analytics
- **Ramakrishna Mission Vivekananda Centenary College** — September 2021 – July 2024
  Bachelor of Science, Mathematics (Honors)

Publication

Real Time Trade Settlement for the US Market, IEEE WCONF 2025. DOI: 10.1109/WCONF64849.2025.11233239